

Exercise. Release week

Q1. Write the Bias-Variance Decomposition equation. How does the phenomenon of overfitting relate to Bias and Variance? What is the indicator of overfitting on a learning curve?

The expected squared error of a model $\hat{f}(x)$ predicting target y is:

$$E[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2$$

where σ^2 is the irreducible noise in the data

Overfitting occurs when a model is overly complex and captures random noise in the training data rather than the true underlying distribution. This corresponds to

→ Low bias: the model can fit the training points almost perfectly

→ High variance: the model's predictions fluctuate widely when given a different training sample.

Overfitting is diagnosed when the training error converges toward zero, while the validation error remains substantially higher and may begin to increase as training progresses.

Q2. Write the formulas for Shannon Entropy and Gini Index. What is the primary difference in the splitting structure between CART and C4.5?

$$\text{Shannon Entropy: } H(S) = - \sum_{i=1}^C p_i \log_2 p_i$$

where p_i is the proportion of samples belonging to class i in node S .

The CART algorithm strictly enforces binary splits at every node. In contrast, the C4.5 algorithm allows for multway splits, creating multiple child nodes simultaneously.

Q3. How are continuous attributes handled for splitting in a Decision Tree? Explain how the C4.5 algorithm handles missing values during training.

The algorithm sorts all unique values of the continuous feature present in the node. It then calculates the midpoints between adjacent distinct values $t = (x_i + x_{i+1})/2$ to use as candidate thresholds. The split is evaluated dynamically as a binary condition: $x < t$ vs. $x \geq t$.

When evaluating a split, instances with missing values are ignored to compute the metric. When routing instances down the tree, C4.5 uses a probabilistic approach: an instance with a missing value is split into fractional instances that are distributed down all child branches weighted by the observed probabilities of the feature's values in the current node.